



College of Engineering
COSC 468: Mobile Programing
Final Project
Spring 2026
Due date: 5th May 2026

Project Title: OpenMarket: Buy & Sell Mobile Application

Introduction:

In this project, you will develop a mobile marketplace application similar to platforms like OpenSooq App, using SwiftUI. The app will allow users to buy and sell products, browse listings, communicate with sellers, and explore nearby items using location-based services. The system will support product listings, user interactions, and transaction management in a user-friendly mobile interface.

Assessed Learning Outcomes:

- **A2:** Apply specialized mobile applications development framework and processes using programming tools.
- **B1:** Design mobile application solutions that solve a real-world problem using appropriate app components and their relationships.
- **B2:** Utilize specialized skills to create formal reports and presentations on mobile application design and implementation.
- **C1:** Operate with responsibility and leadership within a team to achieve the goal of having a mobile application solution to a real-world problem.

Project Description:

This project requires you to build an **OpenMarket mobile application** that enables users to post items for sale, browse available products, communicate with sellers, and complete transactions. The app should include:

1. **Product Listings with Location:** Users can post items with details and location to help buyers find nearby products.
2. **Ratings and Reviews:** Users can rate sellers and provide feedback based on their experience.
3. **Interactive UI/UX:** Create a dynamic and engaging interface that enhances the buying and selling experience.

The app must allow users to:

- Search for products based on category, price, or location.
- Post items for sale with images, descriptions, and prices.
- Browse available items and view detailed product information.
- Contact sellers (e.g., chat or messaging feature).
- Rate and review sellers and transactions.
- View item locations and navigate using maps.
- Add items to favorites or wishlist.

Core Features:

1. Product List View:

- Displays a list of available products.
- Each item includes image, title, price, category, location, and rating.

2. Product Detail View:

- Shows detailed information about the product:
 - Description
 - Price
 - Seller information
 - Ratings and reviews
- Includes options such as:
 - Contact Seller
 - Add to Favorites

3. Add Product (Seller Feature):

- Users can post items for sale by:
 - Uploading images
 - Adding title, description, category, and price
 - Setting location

4. Messaging/Communication:

- Enables buyers to contact sellers directly within the app.

5. Favorites/Wishlist:

- Users can save items for future reference.

6. Location-Based Search:

- Users can find products near their location using GPS.
- Map integration for navigation to item location.

7. Ratings and Reviews:

- Users can rate sellers after transactions.
- Reviews can be written and displayed on seller profiles.

8. User Authentication:

- Users can sign up and log in.
- Personalized features such as saved items, listings, and chat history.

9. Data Persistence:

- Store user data, product listings, messages, and reviews using local storage or Firebase.

Project Scope:

1. UI/UX Design:

- Design an intuitive and visually appealing interface using SwiftUI.
- Include dynamic elements such as product cards, ratings, and real-time updates.

2. Backend Integration:

- Use Firebase or local database to manage:
 - Users
 - Listings
 - Messages
 - Reviews

3. Location Services:

- Integrate GPS to:
 - Show nearby products
 - Display item locations on maps

4. Reviews and Ratings:

- Implement a feedback system for sellers.

5. Marketplace Functionality:

- Allow users to:
 - Post items
 - Browse listings
 - Contact sellers
 - Save favorites

Requirements:

Project Document:

1. Project Overview:

- Summary of the OpenMarket application, objectives, and scope.

2. Technical Specifications:

- Technologies used:
 - SwiftUI
 - Firebase / CoreData
 - CoreLocation
- System architecture and tools.

3. Design Documentation:

- Wireframes and UI mockups:
 - Product listing
 - Product details
 - Add item screen
 - Messaging interface

4. Implementation Details:

- Explanation of features:
 - Product posting
 - Search functionality
 - Messaging
 - Ratings system

5. Testing and Validation:

- Unit testing
- User testing
- Integration testing

6. Challenges and Solutions:

- Issues faced and how they were solved.

7. Future Enhancements:

- Payment integration
- AI-based recommendations
- Advanced filtering
- Push notifications

Xcode Project File:

- **Source Code:** Swift code files for views, data models, and any business logic.

- **Assets:** Images, icons, and other media used in the project.
- **Configuration Files:** Xcode project settings, configurations for backend storage, etc.
- **Dependencies:** If using external libraries (like Firebase or MapKit), include them in the project.

Project Instructions:

1. **Teamwork:**
 - This project will be completed by teams of **up to 3 students**. Each member should focus on a specific feature.
2. **Contribution Documentation:**
 - Each team member must describe their contributions to the project in the **project report**.
3. **Presentation:**
 - Present the project (15-20 minutes) explaining its features, design, and challenges on the presentation date.
4. **Submission:**
 - Submit the **project report** and **Xcode project file** by the due date.

Rubric:

Criteria (CLO)	1–2 (Beginning)	3–4 (Developing)	5–6 (Proficient)	7–8 (Exemplary)	Marks
Software Development Tools (A2)	Struggles to apply mobile development frameworks and tools; limited understanding of programming tools	Demonstrates basic understanding of mobile frameworks and tools with some guidance	Applies appropriate mobile development frameworks and tools independently and effectively	Demonstrates mastery and advanced use of mobile frameworks and programming tools with optimization and best practices	/8
Mobile App Design (B1)	Unable to design a clear mobile application solution; lacks structure and component relationships	Produces a basic design with limited clarity in components and relationships	Designs a functional mobile application with appropriate components and logical relationships	Develops a well-structured, innovative, and scalable design addressing a real-world problem effectively	/8
App Functionality (B1)	Application lacks core functionality or contains major errors	Partial implementation of functionality with noticeable issues	Successfully implements required functionality aligned with the design	Fully functional, robust, and innovative application exceeding expected requirements	/8
Report & Presentation (B2)	Report and presentation are unclear,	Basic report and presentation with limited clarity and organization	Clear, well-structured report and effective presentation of the project	Highly professional report and engaging presentation	/8

Criteria (CLO)	1–2 (Beginning)	3–4 (Developing)	5–6 (Proficient)	7–8 (Exemplary)	Marks
	incomplete, or poorly structured			demonstrating strong communication skills	
Team- work & Leader- ship (C1)	Minimal participa- tion and lack of responsibility within the team	Inconsistent con- tribution with limited collabora- tion	Responsible contri- bution and effec- tive teamwork to- ward project goals	Demonstrates leader- ship, initiative, and strong collaboration to achieve project objectives	/8